

PE-EC702B — Digital Image and Video Processing

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Module 2 Exam Notes: Image Enhancements and Filtering

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1. Gray Level Transformations

Gray level transformations operate directly on individual pixel values in the spatial domain. The basic mapping function is $s = T(r)$, where r is the input intensity and s is the processed intensity.

Standard Gray Level Mapping Functions

1. **Image Negatives:** Reverses the intensity levels of an image:

$$s = L - 1 - r \quad (1)$$

where L is the number of gray levels. Useful for enhancing white or light detail embedded in dark regions.

2. **Log Transformations:** Expands dynamic range of dark pixels while compressing high-level values:

$$s = c \log(1 + r) \quad (2)$$

where c is a constant. Highly useful for displaying Fourier spectra that contain huge intensity ranges.

3. **Power-Law (Gamma) Transformations:** Maps a narrow band of dark values into a wider range:

$$s = cr^\gamma \quad (3)$$

where γ is the exponent.

- $\gamma < 1$: Expands dark regions, compresses bright regions.
- $\gamma > 1$: Compresses dark regions, expands bright regions.

2. Histogram Processing

The histogram of a digital image with intensity levels in $[0, L-1]$ is a discrete function $h(r_k) = n_k$, where n_k is the number of pixels with intensity r_k .

2.1 Histogram Equalization

Histogram equalization is an automated method to improve image contrast by transforming the intensity values such that the output image has a uniform distribution of gray levels.

Continuous Case Derivation

Let r be a continuous variable in $[0, L-1]$. We seek a transformation $s = T(r)$ satisfying:

- (i) $T(r)$ is monotonically increasing (preserves intensity order).
- (ii) $0 \leq T(r) \leq L-1$ for $0 \leq r \leq L-1$.

Using probability theory, the probability density function (PDF) of the transformed variable $s, p_s(s)$, is:

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right| \quad (4)$$

Let us choose the transformation function to be the cumulative distribution function (CDF) of r :

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw \quad (5)$$

Differentiating s with respect to r using the Fundamental Theorem of Calculus:

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = (L - 1)p_r(r) \quad (6)$$

Substituting this back into the PDF expression:

$$p_s(s) = p_r(r) \left| \frac{1}{(L - 1)p_r(r)} \right| = \frac{1}{L - 1} \quad (7)$$

This proves that the resulting PDF $p_s(s)$ is flat (uniform) over the range $[0, L - 1]$.

For discrete images, the transformation function is:

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j) = \frac{L - 1}{MN} \sum_{j=0}^k n_j \quad (8)$$

where MN is the total number of pixels in the image.

2.2 Histogram Specification

Histogram specification allows us to designate the shape of the histogram we wish the processed image to have, rather than just generating a uniform histogram.

1. Equalize the input image to get $s_k = T(r_k)$.
2. Compute the transformation function $G(z_q)$ for the specified output histogram:

$$v_q = G(z_q) = (L - 1) \sum_{i=0}^q p_z(z_i) \quad (9)$$

3. Perform the inverse mapping: $z_q = G^{-1}(s_k)$.

3. Spatial Domain Filtering

Spatial filtering involves moving a filter mask (kernel) over each pixel of an image to perform operations on the pixel neighborhood.

3.1 Smoothing Filters (Low-pass)

- **Linear Smoothing Filters:** Replacing each pixel by the average of its neighbors (e.g., Box filter, Gaussian filter). Reduces noise but blurs edges.

- **Order-Statistics Filters (Non-linear):** Replacement based on ordering (ranking) the pixels in the neighborhood.
 - **Median Filter:** Replaces pixel value with the median of the neighborhood. Excellent for removing **salt-and-pepper (impulse) noise** while preserving edges.
 - **Max/Min Filters:** Replaces with maximum/minimum value (enhances bright/dark details).

3.2 Sharpening Filters (High-pass)

Sharpening highlights transitions in intensity (edges) using spatial differentiation.

1. **First-Derivative Sharpening (Gradient):** The gradient magnitude is used to detect edges. The Sobel operator uses two 3×3 masks to compute horizontal (G_x) and vertical (G_y) gradients:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad (10)$$

2. **Second-Derivative Sharpening (Laplacian):** The isotropic Laplacian operator is defined as:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \quad (11)$$

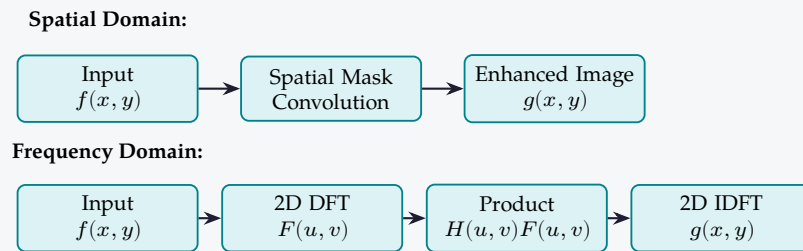
The standard discrete Laplacian mask is:

$$H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{or} \quad H_{\text{diag}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad (12)$$

4. Frequency Domain Filtering

Frequency filtering operates on the 2D Discrete Fourier Transform (DFT) of the image.

Spatial vs. Frequency Domain Filtering Flow



Let $D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$ be the distance from the center of the frequency rectangle.

- **Low-Pass Filters (LPF):** Smooths images by attenuating high frequencies.

- **Ideal LPF:** $H(u, v) = 1$ if $D(u, v) \leq D_0$; else 0. Introduces severe **ringing artifacts** due to sharp transition.
- **Butterworth LPF (order n):** $H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$. Smooth transition, less ringing.
- **Gaussian LPF:** $H(u, v) = e^{-D^2(u, v)/2D_0^2}$. Completely free of ringing.
- **High-Pass Filters (HPF):** Sharpens images (high frequencies are kept, low frequencies attenuated). Formulas are $H_{\text{HPF}}(u, v) = 1 - H_{\text{LPF}}(u, v)$.

5. Worked Examples

Discrete Histogram Equalization

Problem: Equalize a 3-bit image ($L = 8$ levels) of size 64×64 ($MN = 4096$) with the following gray-level distribution:

Gray Level r_k	0	1	2	3	4	5	6	7
Pixel count n_k	790	1023	850	656	329	245	122	81

Solution: We use the transformation formula: $s_k = \text{round} \left((L - 1) \sum_{j=0}^k \frac{n_j}{MN} \right) = \text{round} \left(\frac{7}{4096} \sum_{j=0}^k n_j \right)$. Let's compute the cumulative sum and map the values step-by-step:

1. **For $k = 0$ ($r_0 = 0$):** Cumulative sum $\sum = 790$. $s_0 = \frac{7}{4096} \times 790 = 1.35 \rightarrow \text{round}(1.35) = 1$.
2. **For $k = 1$ ($r_1 = 1$):** Cumulative sum $\sum = 790 + 1023 = 1813$. $s_1 = \frac{7}{4096} \times 1813 = 3.10 \rightarrow \text{round}(3.10) = 3$.
3. **For $k = 2$ ($r_2 = 2$):** Cumulative sum $\sum = 1813 + 850 = 2663$. $s_2 = \frac{7}{4096} \times 2663 = 4.55 \rightarrow \text{round}(4.55) = 5$.
4. **For $k = 3$ ($r_3 = 3$):** Cumulative sum $\sum = 2663 + 656 = 3319$. $s_3 = \frac{7}{4096} \times 3319 = 5.67 \rightarrow \text{round}(5.67) = 6$.
5. **For $k = 4$ ($r_4 = 4$):** Cumulative sum $\sum = 3319 + 329 = 3648$. $s_4 = \frac{7}{4096} \times 3648 = 6.23 \rightarrow \text{round}(6.23) = 6$.
6. **For $k = 5$ ($r_5 = 5$):** Cumulative sum $\sum = 3648 + 245 = 3893$. $s_5 = \frac{7}{4096} \times 3893 = 6.65 \rightarrow \text{round}(6.65) = 7$.
7. **For $k = 6$ ($r_6 = 6$):** Cumulative sum $\sum = 3893 + 122 = 4015$. $s_6 = \frac{7}{4096} \times 4015 = 6.86 \rightarrow \text{round}(6.86) = 7$.
8. **For $k = 7$ ($r_7 = 7$):** Cumulative sum $\sum = 4015 + 81 = 4096$. $s_7 = \frac{7}{4096} \times 4096 = 7.00 \rightarrow \text{round}(7.00) = 7$.

Final Intensity Mapping Summary:

Original r_k	0	1	2	3	4	5	6	7
Mapped s_k	1	3	5	6	6	7	7	7
New Count	790	1023	850	985	0	448	0	0

Note that the output gray levels are now spread out over $[1, 7]$, which significantly enhances contrast.

Quick Revision — Module II

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Key Formulas at a Glance

Image Negative	$s = L - 1 - r$
Power-Law (Gamma)	$s = cr^\gamma$
Histogram Equalization	$s_k = \frac{L-1}{MN} \sum_{j=0}^k n_j$
Laplacian Mask Operator	$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$
2D DFT	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$
Gaussian Low-Pass Filter	$H(u, v) = e^{-D^2(u, v)/2D_0^2}$

One-Line Definitions

- **Median Filtering:** A non-linear spatial enhancement technique that is robust against impulse noise while preserving edges.
- **Isotropic Operator:** A differential operator whose response is independent of the direction of the transitions.
- **Ringling Artifact:** Ripple patterns appearing near sharp boundaries in images filtered with Ideal LPFs due to Gibbs phenomenon.
- **Histogram Equalization:** A method that enhances contrast by transforming the intensity values to achieve a uniform PDF.
- **Sobel Operator:** A first-derivative edge detection operator utilizing gradient masks to compute directional differences.
- **Frequency Domain Filtering:** Modifying the Fourier coefficients of an image before performing the inverse transform.

Frequently Asked / Viva Questions

1. What is the effect of choosing $\gamma < 1$ in power-law transformation?

It expands the dark regions of the image (revealing details in shadows) and compresses the brighter regions.

2. Why does the Ideal LPF produce ringing artifacts while the Gaussian LPF does not?

The transfer function of an Ideal LPF has a sharp discontinuity in the frequency domain, whose Fourier inverse is a sinc function containing spatial oscillations. The Gaussian LPF has smooth Gaussian transitions in both domains, eliminating oscillations.

3. Which filter is best suited for salt-and-pepper noise? Why?

Frequently Asked / Viva Questions (contd.)

The Median filter, because the impulse noise pixels represent outlier values which are completely discarded when the neighborhood values are ranked.

4. **Write down the basic difference between correlation and convolution.**

In spatial correlation, the filter mask is directly multiplied and summed with the image. In convolution, the mask is rotated by 180° before multiplying and summing.

5. **How does the Laplacian filter sharpen an image?**

The Laplacian computes the second derivative, highlighting rapid intensity transitions (edges). By subtracting the Laplacian from the original image ($g(x, y) = f(x, y) - \nabla^2 f(x, y)$), the edges are enhanced.

6. **State the difference between histogram equalization and histogram specification.**

Histogram equalization automatically spreads the intensity levels uniformly, while histogram specification maps the image to a user-defined target histogram shape.

7. **What are Butterworth filters?**

Butterworth filters have a transfer function with a transition band whose sharpness is controlled by the order parameter n , striking a balance between Ideal LPF (sharp transition) and Gaussian LPF (very smooth transition).

8. **Why do we shift the origin of the 2D DFT before filtering?**

Shifting the origin (using $f(x, y)(-1)^{x+y}$) centers the low-frequency component $F(0, 0)$ at the center of the frequency rectangle, simplifying filter design.

9. **Define the gradient of an image $f(x, y)$.**

The gradient is a 2D vector $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]^T$ that points in the direction of the maximum rate of change of intensity.

10. **What is frequency domain high-pass filtering?**

HPF attenuates the low frequencies (representing smooth variations) while passing the high frequencies (representing edges and noise), resulting in sharpened details.

Comparison of Sharpening Operators

Attribute	First-Derivative (Gradient)	Second-Derivative (Laplacian)
Response at Constant	Zero response	Zero response
Response at Step/Ramp	Non-zero along the ramp	Non-zero only at start/end of ramp
Ramp Response Width	Produces thick, double-pixel edges	Produces thin, single-pixel edges